

DEEKSHITHA LAKSHMI KALAVALA

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Professional Summary

AWS SAA-C03 & Azure AZ-400 certified Cloud & MLOps Engineer with 1+ year of production experience deploying end-to-end ML pipelines, CI/CD systems, and containerized microservices on Azure and AWS. Author of a peer-reviewed IEEE publication on ML fairness in healthcare diagnostics. Delivered 99.9% uptime, 30% latency reduction, and 40% faster release cycles in industry. Passionate about building reliable, scalable cloud-native systems in healthcare and enterprise environments.

Education

University of Cincinnati

Master of Engineering in Computer Science | GPA: 3.8/4.0

- **Relevant Coursework:** Distributed Systems, Cloud Computing, ML Systems, Algorithms & Data Structures, Database Systems

KL University

Bachelor of Technology, Computer Science & Engineering | GPA: 9.12/10

Cincinnati, OH

Aug 2024 – May 2026

Andhra Pradesh, India

Jan 2019 – Mar 2023

Skills

Programming: Python, Java, SQL, Bash, JavaScript

Cloud & DevOps: AWS (EC2, S3, IAM, Lambda, CloudWatch), Azure (AZ-900, AZ-400, Entra ID, VMs, Functions, Blob Storage, Container Apps), Docker, Kubernetes, Terraform, Helm, ArgoCD

Cloud Architecture: IaaS, PaaS, SaaS, Virtualization, Elastic Scalability, Solutions Architecture, Network Management, Cloud Security, Hybrid Cloud

CI/CD & Automation: GitHub Actions, Jenkins, Azure DevOps Pipelines, Blue-Green Deployments

Backend & APIs: Spring Boot, FastAPI, Django, REST APIs, Microservices

ML & Data: Scikit-learn, XGBoost, MLflow, Azure ML SDK, SMOTE, PCA, Feature Engineering, Pandas, NumPy

Databases: PostgreSQL, MySQL, MongoDB, Redis

Visualization & Monitoring: Power BI, Streamlit, Folium, Grafana, Azure Monitor, Matplotlib

Security & Reliability: IAM, Secrets Management, SLA/SLO, Health Checks, Incident Response

Software Practices: Agile/Scrum, Unit Testing, System Testing, Technical Documentation, Change Control, Troubleshooting, Git, Code Reviews, Postman, Jira, Linux

System Design: Distributed Systems, High-Availability Architecture, Scalability, Fault Tolerance

Experience

KBK Business Solutions

Software Developer | Java, Spring Boot, Azure, Jenkins, PostgreSQL

Hyderabad, India

Jul 2023 – Mar 2024

- **Reduced API p95 latency by 30%** (from ~400ms to ~280ms) by introducing Redis caching and optimized SQL query execution on Spring Boot microservices serving 10K+ daily requests.
- **Eliminated manual release steps** by engineering GitHub Actions + Jenkins CI/CD pipelines for automated build, unit testing, and production deployment, cutting release cycle from 2 weeks to 3 days for a 5-engineer team and reducing deployment errors by 90%.
- **Achieved 99.9% uptime** across 3 Azure-hosted microservices by implementing health checks, auto-restart policies, blue-green deployment strategies, and incident response protocols; authored technical documentation and change control runbooks for operational procedures.
- Troubleshoot and resolved production errors across microservices, **improving data retrieval speed by 25%** via PostgreSQL indexed views and query plan optimization for dashboards serving 50+ stakeholders.
- Designed and implemented **solutions architecture** for hybrid cloud deployment across Azure PaaS and IaaS services, incorporating virtualization, containerization (Docker), and elastic scalability patterns to support fluctuating workloads.
- Collaborated cross-functionally with 3 teams to deliver system testing, document testing outcomes, and prepare complete technical documentation aligned with the software development lifecycle (SDLC) and change control standards.
- Monitored system performance and functionality using **Azure Monitor and Grafana**, proactively identifying bottlenecks and gathering data to inform future development feasibility studies.

Indian Servers – Software Development Company

Software Engineering Intern | Python, ML, Automation

Vijayawada, India

Jun 2021 – Aug 2021

- Built Python automation and unit testing scripts, **reducing QA regression effort by 20%** across 200+ test cases; supported system testing, documented outcomes, and contributed to technical documentation for 3 product modules.
- Engineered an ML classification pipeline with feature engineering and cross-validation, **achieving 87% accuracy** on a real-world imbalanced dataset using Scikit-learn — demonstrated understanding of application development lifecycle from prototyping to evaluation.
- Participated in troubleshooting and deciphering error messages for an internal web application, identifying root causes and escalating resolution plans to senior engineers, improving team problem-resolution response time by 30%.
- Contributed to user and technical documentation for internal tools, applying SDLC processes and operating procedures to maintain system consistency and support knowledge transfer across the team.

Projects

Biomedical Class Imbalance – IEEE Published Research | *Python, PCA, Near Miss, SVM, Scikit-learn* [IEEE Xplore](#)

- Published peer-reviewed research in IEEE Xplore applying PCA + Near Miss undersampling to 5 biomedical datasets; **improved minority-class recall by 22%** over baseline SVM, Random Forest, and Logistic Regression — addressing real-world ML fairness in healthcare diagnostics and clinical data management.
- Designed complete experimental methodology including data ingestion, feature extraction, model training, validation, and documentation of testing outcomes across multiple classification algorithms.
- Demonstrated analytical skills by identifying system performance gaps in existing ML approaches and developing technical solutions to improve clinical diagnostic accuracy on imbalanced real-world datasets.

Azure MLOps Pipeline – Customer Churn Prediction | *Python, Azure ML, FastAPI, Docker, GitHub Actions* [GitHub](#)

- Architected an end-to-end MLOps pipeline on Azure ML (PaaS): automated data ingestion, model training, versioning with MLflow, and deployment to a **managed online endpoint serving real-time inference** — implementing full solutions architecture across cloud services.
- Containerized a FastAPI inference API with Docker and **automated retraining + production deployment via GitHub Actions CI/CD**, achieving zero-touch releases on every code push with built-in unit testing gates.
- Trained an XGBoost churn classifier; applied SMOTE and hyperparameter tuning to **boost F1-score by 18%** over baseline Logistic Regression; monitored model performance and system functionality post-deployment using Azure Monitor.
- Documented complete technical architecture, deployment runbooks, and change control procedures to support ongoing system maintenance and future development feasibility analysis.

NYC Real Estate Valuation – Location-Aware ML System | *Python, XGBoost, KMeans, Streamlit, Folium* [GitHub](#)

- Compared **5 ML models** on NYC housing dataset; engineered spatial features via **Haversine distance** and **KMeans clustering (k=10)**, with XGBoost achieving the highest R^2 score — demonstrating strong analytical and technical problem-solving skills.
- Delivered a **multi-page Streamlit dashboard** with interactive Folium heatmaps, EDA visualizations, model comparison charts, feature importance plots, and a real-time price predictor for end users.
- Designed and prepared complete user documentation and technical documentation covering system architecture, data pipeline design, model evaluation methodology, and deployment instructions.

Certifications & Achievements

Cloud: AWS Certified Solutions Architect – Associate (SAA-C03), Microsoft Azure DevOps Engineer Expert (AZ-400), Azure Fundamentals (AZ-900), Oracle Cloud Infrastructure Foundations

ML & Dev: Building Cloud Services with Java Spring Framework, Django for Everybody Specialization, AI Foundations: Machine Learning, Database Management Essentials

Competitive Coding: 100–200 LeetCode problems | Focus: Arrays, Trees, Graphs, Dynamic Programming

Languages: English (Professional), Telugu (Native), Hindi (Working)